RoleWalk: inductive structural roles for directed graphs from the characteristic function of random walks

Maixent Chenebaux

Draft of 01 November 2026 — results and proofs; prose to come

Abstract. Structural roles group nodes by their position in a graph rather than by proximity. RoleWalk describes each node by the characteristic function of its random-walk transition probabilities, sampled at a few frequencies, together with the distribution of the walk's mass over distance layers and, optionally, low-order moments of a probabilistic curvature. The embedding is local, deterministic, reads directions and weights, and needs no training. We prove that it is invariant and local (hence transferable across graphs), that its curvature-free distances are maximum mean discrepancies, that its characteristic-function block is Lipschitz in edge insertions, that it recovers triangle counts on regular graphs, so that it separates graphs 1-WL cannot (and conversely: the two are incomparable), and that distance-regular graphs with equal intersection arrays are a provable failure class. With a budget on the probability mass that pruning may drop, the approximation error is certified. Empirically it matches or beats struc2vec, GraphWave, Role2Vec, RESD and ReFeX on role classification at a fraction of the cost of every method but ReFeX, is best on real directed roles and on transfer of centralities across graphs, embeds a million nodes in 20 s to 2.5 min on one core, and shows a preliminary improvement of GPS on ZINC when combined with RWSE.

1 Introduction

Claims, each backed by a result below:

1. All of: inductive, directed, weighted, training-free, deterministic, online, fast (Table 1).
2. Accuracy at or above the state of the art on role classification at a fraction of the cost; a million nodes in minutes on one core (Table 2, Figure 1, Table 11).
3. Reads edge directions natively, including walks that change direction; best on real directed roles and on unsupervised synthetic ones (Table 4, Table 5, Proposition 10).
4. Pruning with a certified error (Proposition 13).
5. Transfer: a regressor fitted on one graph predicts path-based centralities (betweenness, closeness, harmonic) of another better than every baseline (Table 8; Proposition 1).
6. Separates graphs that 1-WL cannot with 3 walk steps (the two are incomparable), close to 16-step RWSE on BREC; a provable failure class (Section 4, Table 6).
7. Useful as a structural encoding for graph transformers (Table 10).

2 Related work

Feature-based roles: ReFeX [1], RolX [2]. Walks and word2vec: struc2vec [3], Role2Vec [4]; directed and weighted extensions of struc2vec [5] [TODO: compare if code is available]. Diffusion: GraphWave [6], closest to RoleWalk (Proposition 11). Deep role embeddings: RESD and the RONE benchmark [7] [TODO: exact references]. Structural encodings for GNNs: RWSE [8], used by GPS [9]. Expressiveness: 1-WL [10], [11], [12], CSL [13], BREC [14]. Curvature: Ollivier-Ricci [15], [16]. Kernel mean embeddings and MMD: [17].

3 Method

Walk. $G = (V, E)$ with non-negative weights A (directed or not), $\tilde{D} = \mathrm{diag}((I + A)\mathbf{1})$ and the self-loop-augmented walk $H = \tilde{D}^{-1}(I + A)$. For a node u and a step w, $p_u^w = e_u^\top H^w$ is the distribution of the walk after w steps. Directed graphs use the out-walk H_+ (A) and the in-walk H_- ($A^\top$) and concatenate both embeddings (138 features, 150 with curvature). As an option, the characteristic functions of the rows of

H_+H_- (one step forward, one back: nodes sharing targets with u) and H_-H_+ (shared sources) add $4m = 40$ features (Proposition 10).

Characteristic function. With $f_\theta(x) = e^{i\theta x} - 1$ and a grid $\Theta = \{\theta_1, ..., \theta_m\}$ (log-spaced in $[1, 100]$, $m = 10$),

$$\varphi_u^w(\theta) = \sum_{j \in V} f_\theta\big(p_{uj}^w\big), \quad \theta \in \Theta. \tag{1}$$

Since $f_\theta(0) = 0$, the sum runs over the support of p_u^w only.

Onion layers. With $d(u, j)$ the length of the shortest walk from u to j, $s_u^w(k) = \sum_{j:d(u,j)=k} p_{uj}^w$ for $k = 0, ..., w$: the mass the walk keeps at each distance, which the multiset of probabilities ignores. Supports of earlier powers give the layers, so no extra walk is computed.

Curvature (option). For an edge xy of the symmetrised unweighted graph, $\kappa(x, y) = \sum_z \min\big(m_x(z), m_y(z)\big)$ with m_x the one-step lazy walk from x; κ_x is its mean over the edges at x. Each step adds the mean $\mu_u^w = \sum_j p_{uj}^w \kappa_j$ and the standard deviation σ_u^w of κ under p_u^w. Since a unit of mass that does not overlap moves at least one step, $W_1\big(m_x, m_y\big) \geq 1 - \kappa(x, y)$: κ is an upper bound of Ollivier-Ricci curvature $1 - W_1$, made of probabilities.

One measure, one kernel. Every node j reached at step w is an atom marked by its step and its layer, which gives the *footprint* of the walks from u:

$$\mu_u = \sum_{w=1}^{L} \sum_{j: p_{uj}^w > 0} \delta_{\left(w, p_{uj}^w, d(u,j)\right)}. \tag{2}$$

The embedding is the integral of a single feature map over the footprint (a finite measure, not normalised), $X_u = \int \Psi \, d\mu_u$, with

$$\Psi(w, a, \ell) = e_w \otimes (\psi(a) \oplus \|\Theta\| a e_\ell), \quad \psi(a) = \big(f_\theta(a)\big)_{\theta \in \Theta}, \tag{3}$$

which gives, for each step $w = 1, ..., L$ ($L = 3$), $\operatorname{Re}\varphi_u^w$, $\operatorname{Im}\varphi_u^w$ and $\|\Theta\| s_u^w$. With curvature the mark is $\big(d(u, j), \kappa_j\big)$ and Ψ adds $\|\Theta\| a(\kappa, \kappa^2)$; we report $\|\Theta\|(\mu_u^w, \sigma_u^w)$, which carries the same information but, being nonlinear, is not an integral of Ψ: kernel statements below are for the curvature-free embedding (or raw moments). In all, $2mL + L(L + 3)/2(+2L) = 69$ (75) features. The probability block ignores the layer; the layer block is the first-order part of one characteristic function per layer, which is what fixes the factor $\|\Theta\| = \big(\sum_\theta \theta^2\big)^{1/2}$ (Proposition 3). Without it the layers carry 0.02% of the variance and do not change the roles found by k-means (Section 6.4).

Pruning. After each product, entries below $\text{tol} = 10^{-4}$ are dropped; their mass is added back to $\operatorname{Im}\varphi$ to first order ($\theta \cdot \text{mass}$), the same tangent approximation as the layer block. The first step is never pruned. Dropped mass can gather again: in a star with more than 10^4 leaves, the leaf mass of step 2 falls below tol although half of it returns to the centre at step 3. With a budget δ (`max_missing`), a row drops its small entries only while its missing mass $r_u^w = 1 - \|q_u^w\|_1$ stays below δ, which bounds the error (Proposition 13). Distances are read from supports before pruning: for $L \leq 3$ every retained entry gets its exact distance, while layer masses still lose what pruning drops. A row whose small entries would exceed the budget keeps all of them. Results in Section 4 are otherwise stated for $\text{tol} = 0$.

Implementation. For each root, one pass propagates a sparse row with a dense accumulator, prunes it, and summarises it (characteristic function, layers from the step at which each node is first reached, curvature moments) without storing the powers: $O(n)$ memory per thread, and rows are independent. Below $p = 10^{-3}$ every angle θp is under 0.1, and the characteristic function comes from the power sums $\sum p^k$, $k \leq 8$ instead of $2m$ sines (truncation error below $\theta^9 p_{\max}^8 / 9! < 3 \cdot 10^{-12}$ per coordinate and row in exact arithmetic; outputs are float32).

Roles. Ward clustering with silhouette for the number of roles, or k-means with a given number; fitted centroids label nodes of any graph.

4 Properties

Throughout, L is the number of steps, $B_r(u)$ the subgraph induced by nodes within distance r of u (out- and in-distance for directed graphs; with curvature or mixed probes, distance in the underlying undirected graph, which contains everything these read), graphs have finite non-negative weights and no self-loops, distances are taken in the graph of positive-weight edges, m_x and κ are defined on the simple unweighted symmetrisation ($\kappa = 0$ at isolated nodes), and statements hold for tol $= 0$, in exact arithmetic. Mixed probes read the undirected 2-ball, so with them the locality radius is $\max(L, 2)$.

Proposition 1 (locality and invariance)**.** *X_u is determined by the isomorphism class of the rooted graph $(B_L(u), u)$, and of $(B_{L+2}(u), u)$ with curvature. In particular it does not depend on $|V|$, nodes related by an automorphism get equal embeddings, and a node can be embedded from its neighbourhood alone.*

Proof. $p^w_{uj} = \sum_{u=x_0,\dots,x_w=j} \prod_t H_{x_t x_{t+1}}$ runs over walks of length $w \le L$ from u. Their nodes lie within distance w, and $H_{xy} = (I + A)_{xy}/\tilde{D}_{xx}$ needs the edges at x, with x within distance $w - 1$: all are in $B_L(u)$. An isomorphism of rooted balls maps these walks bijectively and preserves their weights, so it maps p^w_u to $p'^w_{u'}$ entry by entry; Equation 1 is a sum over j, hence equal. Distances up to L are read inside $B_L(u)$. With curvature, κ_j for j within distance L needs the closed neighbourhoods of j and of its neighbours, inside $B_{L+2}(u)$. □

Coordinates therefore mean the same thing in every graph, which is what transfer (Table 8, Table 9) requires.

Proposition 2 (injectivity)**.** *Let P, Q be finite multisets of reals in $(0, 1]$. If $\sum_{p\in P} f_\theta(p) = \sum_{q\in Q} f_\theta(q)$ for all θ in a set with an accumulation point (e.g. $[1, 100]$), then $P = Q$. For a finite grid, given any countable family of pairs $P \ne Q$, Lebesgue-almost every $\Theta \in \mathbb{R}^m$ separates all of them.*

Proof. $F(\theta) = \sum_P e^{i\theta p} - \sum_Q e^{i\theta q} - (|P| - |Q|)$ is entire and vanishes on a set with an accumulation point, hence everywhere. Exponentials $e^{i\theta\lambda}$ with distinct λ are linearly independent, so each $\lambda > 0$ has the same multiplicity in P and Q (and $|P| = |Q|$, coefficient of $\lambda = 0$). For the second claim, the real zeros of each nonzero entire function F are countable; a countable union of countable sets is null, so almost every θ_1 already separates every pair. □

So with Θ an interval, φ^w_u is exactly the multiset of nonzero transition probabilities of u after w steps. The second claim is qualitative: the separation margin, not its existence, motivates $m = 10$ frequencies across two decades.

Proposition 3 (one kernel on the footprint)**.** *Without curvature, $\|X_u - X_v\|^2 = \mathrm{MMD}^2_K(\mu_u, \mu_v)$ (extended to finite measures) for the positive-semidefinite kernel on marked atoms*

$$K((w, a, \ell), (w', b, \ell')) = \delta_{ww'}\big[k_\Theta(a, b) + \|\Theta\|^2 ab\,\delta_{\ell\ell'}\big], \tag{4}$$

$$k_\Theta(a, b) = \sum_{\theta\in\Theta} [\cos\theta(a - b) - \cos\theta a - \cos\theta b + 1]. \tag{5}$$

The first term compares probabilities whatever their layer, the second compares masses in the same layer. Moreover, with $t = \max(\Theta)\max(a, b)$,

$$\big|k_\Theta(a, b) - \|\Theta\|^2 ab\big| \le \|\Theta\|^2 ab\, t^2(7/12 + t^2/36), \tag{6}$$

so the layer term is the first-order part of $k_\Theta(a, b)\,\delta_{\ell\ell'}$, the kernel of one characteristic function per layer, and $\|\Theta\|$ is the only factor for which the two agree to first order.

Proof. $X_u = \sum \Psi$ over the atoms of μ_u, so X_u is the kernel mean embedding of μ_u and $\|X_u - X_v\|^2$ is the MMD [17] for $K = \mathrm{Re}\langle\Psi, \Psi'\rangle$, positive semidefinite as an inner product. The blocks e_w and e_ℓ are orthonormal, which gives $\delta_{ww'}$ and $\delta_{\ell\ell'}$; expanding $\mathrm{Re}\big(e^{i\theta a} - 1\big)\overline{(e^{i\theta b} - 1)}$ gives k_Θ. Unreached nodes have $\Psi = 0$ and do not count. For the bound, fix θ and write the real part as $(\cos\theta a - 1)(\cos\theta b - 1) + \sin\theta a \sin\theta b$. Since $|\cos x - 1| \le x^2/2$, the first product is at most $\theta^4 a^2 b^2/4 \le \theta^2 ab\, t^2/4$. With $\sin\theta a = \theta a + r_a$ and $|r_a| \le \theta^3 a^3/6$, $\big|\sin\theta a \sin\theta b - \theta^2 ab\big| \le \theta^4 ab(a^2 + b^2)/6 + \theta^6 a^3 b^3/36 \le \theta^2 ab\, t^2(1/3 + t^2/36)$. Summing over θ gives the bound, and $\|\Theta\|^2 = \sum_\theta \theta^2$ is the coefficient of ab. □

The concatenation is therefore a sum of kernels on one object: a mark-blind term on probabilities and a mark-aware term on masses. The first-order regime $t \ll 1$ holds where walks spread thin, not for low-degree

nodes nor for returns to a hub ($\max(\Theta) = 100$): the linear layer kernel is a choice, $L(L+3)/2 = 9$ coordinates instead of $mL(L+3) = 180$ for one characteristic function per layer, and the full per-layer kernel is not better (Table 13). Further marks enter the same way, as factors of the second term: curvature, or labels on the steps of each path.

Proposition 4 (stability)**.** *Let G be simple, undirected and unweighted, and G' be G plus an edge ab, with $d = \min(d_a, d_b)$ the smaller endpoint degree in G. For every node u and step w,*

$$\left\|\varphi_u^w(G') - \varphi_u^w(G)\right\|_2 \le w\|\Theta\|_2 \cdot \frac{2}{d+2}, \tag{7}$$

and the left side is 0 if u is at distance $\ge w$ from both a and b.

Proof. $|e^{i\theta x} - e^{i\theta y}| \le \theta\,|x-y|$, so $\left\|\varphi_u^w(G') - \varphi_u^w(G)\right\|_2 \le \|\Theta\|_2 \|p_u'^w - p_u^w\|_1$. Telescoping, $H'^w - H^w = \sum_{t=0}^{w-1} H'^t(H'-H)H^{w-1-t}$; stochastic matrices have $\|\cdot\|_\infty = 1$, so each row of the difference has ℓ_1 norm at most $w\|H'-H\|_\infty$. Only rows a and b of H change; row a goes from d_a+1 entries $1/(d_a+1)$ to d_a+2 entries $1/(d_a+2)$, an ℓ_1 change of $2/(d_a+2)$. Row u of $H'^t(H'-H)$ is nonzero only if a walk of $t \le w-1$ steps from u reaches a or b. □

Insertions at high-degree nodes barely move the characteristic-function block, and the effect is confined to the $(L-1)$-ball. On 200 random insertions (powerlaw-cluster graphs, 60 nodes) the largest observed move of this block is 0.52 of the bound (`theory_checks.py`). The bound does not extend to the layer block, which reads distances: in $K_{2,d}$, joining the two hubs a, b moves $s_a^2(2)$ from $d/(3(d+1))$ to 0, whatever d.

Proposition 5 (degree and triangles)**.** *Let G be simple, undirected and unweighted. Then $s_u^1(0) = 1/(d_u+1)$, and if G is d-regular ($d > 0$), $s_u^3(0) = (H^3)_{uu} = (1+3d+2t_u)/(d+1)^3$, with t_u the number of triangles at u. Hence RoleWalk with $L \ge 3$ separates 1-WL-equivalent graphs, e.g. two triangles and a hexagon, or any two d-regular graphs with different triangle counts.*

Proof. $(I+A)^3 = I + 3A + 3A^2 + A^3$ with $(A)_{uu} = 0$, $(A^2)_{uu} = d$, $(A^3)_{uu} = 2t_u$, and $H = (I+A)/(d+1)$. 1-WL gives every node of two d-regular graphs of the same size the same colour. □

Proposition 6 (relation to RWSE)**.** *The distance-0 layer $s_u^w(0) = (H^w)_{uu}$ is the return probability of the lazy walk, i.e. RWSE [8] for H, up to L steps. On a d-regular graph, $\left\{(H^w)_{uu}\right\}_{w \le L}$ and the simple-walk RWSE $\left\{(P^k)_{uu}\right\}_{k \le L}$, $P = A/d$, determine each other.*

Proof. $H = (I+dP)/(d+1)$, so $(H^w)_{uu} = (d+1)^{-w}\sum_{k=0}^{w}\binom{w}{k} d^k (P^k)_{uu}$: a triangular system with nonzero diagonal $d^w/(d+1)^w$. □

On irregular graphs the binomial transformation above does not apply; RoleWalk adds the other layers and the full distribution of transition probabilities.

Proposition 7 (a failure class: distance-regular graphs)**.** *Let G, G' be distance-regular graphs with the same intersection array. Then all nodes of G and G' get the same embedding, for every L, with or without curvature. This covers strongly regular graphs with equal parameters (diameter 2).*

Proof. Let A_i be the distance-i matrices. Distance-regularity gives $AA_i = b_{i-1}A_{i-1} + a_iA_i + c_{i+1}A_{i+1}$ with coefficients from the intersection array [18], so by induction $(I+A)^w = \sum_i \alpha_{w,i}A_i$ with $\alpha_{w,i}$ depending only on the array. As $H = (I+A)/(k+1)$, $p_{uj}^w = \alpha_{w,d(u,j)}/(k+1)^w$: each row has value $\alpha_{w,i}/(k+1)^w$ with multiplicity k_i, the number of nodes at distance i, fixed by the array. So φ_u^w and $s_u^w(i) = k_i\alpha_{w,i}/(k+1)^w$ are the same for all nodes of both graphs. For an edge xy, $\kappa(x,y) = |N[x] \cap N[y]|\ /(k+1) = (a_1+2)/(k+1)$, a constant, so μ_u^w is constant and $\sigma_u^w = 0$. □

Checked on Shrikhande / 4×4 rook, Chang / $T(8)$ and the diameter-3 pair Doob $D(1,1)$ / Hamming $H(3,4)$: largest difference between any two nodes $\le 3 \cdot 10^{-7}$ (float32). This explains the 0/20 distance-regular pairs of BREC (Table 7); RWSE, which is constant on these graphs for the same reason, fails them too.

Proposition 8 (locality limits graph-level separation)**.** *Let A and B be a 40-cycle with pendant nodes attached at cycle positions $\{0, 15\}$ and $\{0, 20\}$. They are not isomorphic and 1-WL separates them, but RoleWalk with $L = 3$ (with curvature) does not.*

Proof. The pendants are more than $2(L+2)$ apart in both graphs, so every rooted ball $B_{L+2}(u)$ contains at most one pendant and is fixed by u's distance to it; both graphs have the same multiset of balls. Conclude by Proposition 1. For 1-WL, let h mark the two degree-3 nodes, which round 1 identifies; round $t+1$ colours

determine $(A^t h)_v$, the number of t-step walks from v to a marked node. Both graphs are bipartite (an even cycle with pendant trees). In A, the only 15-step walk from node 0 to node 15 is the short arc, and no odd walk returns to 0, so $(A^{15}h)_0 = 1$; in B, nodes 0 and 20 are on the same side, so $(A^{15}h)_0 = 0$. After 16 rounds the marked nodes have different colours. □

Proposition 9 (RoleWalk and 1-WL are incomparable)**.** *Proposition 5 gives graphs that RoleWalk separates and 1-WL does not. Conversely, without curvature, for $L \in \{2, 3\}$ there are nodes that L-round 1-WL (uniform initial colours) separates and RoleWalk with L steps does not, for every set of frequencies.*

Proof. Take a clique K_5 on $\{r, a, b, c, d\}$ and ten further nodes, adjacent to b, c, d in G, and five to a, b and five to c, d in G'. The neighbour degrees of r are $\{4, 14, 14, 14\}$ in G and $\{9, 9, 9, 9\}$ in G', so two rounds separate r. Walks from r agree: at step 1 each clique node has probability $1/5$; at step 2 each clique node has $1/5(1/5 + 2/5) = 3/25$ (the inverse augmented degrees of the neighbours of r sum to $1/5 + 3/15 = 2/5$ in G and $4/10 = 2/5$ in G') and each outer node $1/25 = 1/5 \cdot 3/15 = 1/5 \cdot 2/10$, with the same layers. For $L = 3$, add a leaf z at r and root at z: three rounds separate z. From z, step 1 puts $1/2$ on z and r; step 2 puts $1/3$ on z and r and $1/12$ on each of a, b, c, d; step 3 puts $2/9$ on z, $23/90$ on r, $4/45$ on each of a, b, c, d and $1/60$ on each outer node, in both graphs, with layers $(2/9, 23/90, 16/45, 1/6)$ (exact rational computation, `theory_checks.py`). With curvature, both pairs are separated. These are node-level collisions; Proposition 8 gives a graph-level one. □

On the 6,781 nodes of the 996 connected graphs with at most 7 nodes, no pair separated by 3-round 1-WL is confused by RoleWalk ($L = 3$), which splits 571 of the 1-WL classes (embeddings grouped at a sup-distance below 10^{-4} of the largest coordinate; `theory_checks.py`); the counterexamples above have 15 and 16 nodes. RoleWalk is also local, which 1-WL with unbounded rounds is not.

Proposition 10 (walks that change direction)**.** *Let u be a source pointing to two sinks (G), or one of the two sources of a directed $K_{2,2}$ (G'). For every L, with or without curvature, u gets the same out- and in-walk embedding in G and G'; but $e_u^\top H_+ H_-$ has values $(2/3, 1/6, 1/6)$ in G and $(5/9, 2/9, 1/9, 1/9)$ in G'; at $\theta = 1$ (in the default grid) $\operatorname{Re}\varphi$ is -0.2418 and -0.1873, so the mixed probes separate them.*

Proof. Out-walk: u keeps $1/3$ and sends $1/3$ to each sink, where it stays; the layers agree. In-walk: u has no in-edge and stays. The symmetrised graphs are a path $a - u - b$ and a 4-cycle; no edge has a common neighbour, and every edge has $\max(d_x, d_y) = 2$, so every node has $\kappa = 2/3$ in both. In H_-, a sink with in-neighbours N spreads uniformly over itself and N: $1/2$ on itself and u in G, $1/3$ on itself, u and the other source in G'. Hence $e_u^\top H_+ H_- = 1/3\big(e_u + \sum_{\text{sinks } s} e_s^\top H_-\big)$, which gives the two rows. □

The out- and in-walks see where u sends and receives, not who shares its targets (voters of the same candidates) or its sources; Table 4 and Table 5 measure what this adds.

Proposition 11 (relation to GraphWave)**.** *GraphWave [6] embeds u by $\chi_u(t) = \frac{1}{n}\sum_j e^{it\Psi_{ju}}$ with $\Psi = U g_s(\Lambda) U^\top$ the heat kernel at a scale s chosen from the spectrum of each graph. Then $n(\chi_u(t) - 1) = \sum_j f_t(\Psi_{ju})$: Equation 1 with the heat kernel in place of H^w. χ_u depends on n and on s, so the same rooted neighbourhood can get different coordinates in graphs of different sizes or spectra.*

Proof. $\sum_j \big(e^{it\Psi_{ju}} - 1\big) = n\chi_u(t) - n$. Ψ has full support on a connected graph, so Proposition 1 fails; $1/n$ and s are graph-level. □

This explains most of why GraphWave transfers at chance level: without its $1/n$, it reaches 0.455 on airport transfer, against 0.488 for RoleWalk (Table 9).

Proposition 12 (cost)**.** *With threshold pruning (no budget), each row of each power after the first keeps at most $1/\operatorname{tol}$ entries (the first step keeps all $d_u + 1$; a budget can keep any number), so step w costs $O\Big(\sum_u \sum_{j \in \operatorname{supp} p_u^{w-1}} \big(d_j + 1\big)\Big)$ with $|\operatorname{supp} p_u^{w-1}| \le \min(|B_{w-1}(u)|, 1/\operatorname{tol})$; the characteristic function adds $O(m \cdot \operatorname{nnz})$, the layers $O(\operatorname{nnz})$. As $\kappa(x, y) = \big(c_{xy} + 2\big)/\big(\max(d_x, d_y) + 1\big)$ with c_{xy} the common neighbours of x and y, curvature costs a triangle enumeration, $O\big(|E|^{3/2}\big)$. Embedding a set S touches only $B_L(S)$ ($B_{L+2}(S)$ with curvature).*

Proof. Entries are non-negative and each row sums to at most 1, so at most $1/\operatorname{tol}$ of them are $\ge \operatorname{tol}$. Both closed neighbourhoods are uniform ($1/(d+1)$ on $d+1$ nodes) and share x, y and their common neighbours, so the overlap is $c_{xy} + 2$ times the smaller mass; orienting edges by degree enumerates each triangle once in $O\big(|E|^{3/2}\big)$. The rest follows from sparse products and Proposition 1. □

Proposition 13 (certified pruning)**.** *For one walk direction, without curvature or mixed probes, $L \le 3$ and in exact arithmetic, let $\hat{X}_u$ be computed with pruning and r_u^w the mass row u misses at step w. Then*

$$\left\| X_u - \hat{X}_u \right\|_2 \le \sqrt{5}\|\Theta\|_2 \left(\sum_{w \le L} (r_u^w)^2 \right)^{1/2}, \tag{8}$$

so with a budget δ, $\left\| X_u - \hat{X}_u \right\|_2 \le \sqrt{5L}\|\Theta\|_2 \delta$ ($\sqrt{10L}\|\Theta\|_2\delta$ for the two directions of a directed graph). Mixed probes with missing masses r_β add $4\|\Theta\|_2^2 \sum_\beta r_\beta^2$ to the squared bound.

Proof. Propagation and deletion keep entries non-negative, so the pruned row q_u^w satisfies $0 \le q_u^w \le p_u^w$ entrywise and $\|p_u^w - q_u^w\|_1 = r_u^w$. For each θ, $\left|\sum_j f_\theta(p_j) - f_\theta(q_j)\right| \le \theta r_u^w$, and the first-order correction $i\theta r_u^w$ adds at most θr_u^w: the characteristic-function block moves by at most $2\|\Theta\|_2 r_u^w$. Layers use the exact distances (supports read before pruning are exact up to step 2, and every node reached at step 3 outside them is at distance 3), so each layer loses $\sum_{j \in \text{ layer}} (p_j - q_j) \ge 0$, in total r_u^w: the scaled layer block moves by at most $\|\Theta\|_2 r_u^w$. Summing squares over blocks and steps gives the bound; a budget enforces $r_u^w \le \delta$, since propagation by a stochastic matrix keeps the missing mass and a row only drops entries while it stays below δ. Directions are orthogonal blocks; a mixed probe is a characteristic function without layers, hence $2\|\Theta\|_2 r_\beta$. □

The bound is a posteriori without a budget ($r_u^w \le 1$) and a priori with one; it is absolute in feature space ($\|\Theta\|_2 \approx 125$, so about 24 at $\delta = 0.05$, $L = 3$), not a guarantee on downstream accuracy. Floating-point evaluation adds its own error.

5 Experiments

All CPU timings on one core. Classification: logistic regression on standardised embeddings, 50 stratified 80/20 splits. Best in bold, second underlined, among embeddings; the degree baseline (log-degree alone, which shows how far labels follow activity) is set apart and not ranked. Code and CSV files in `benchmarks/`.

Table 1: Properties, checked against each paper: ✓ yes, ✗ no, ~ partial.

	Local, fixed coords	Directed	Weighted	No training	Online	Deterministic	Time (USA)
RoleWalk	✓	✓	✓	✓	✓	✓	0.2 s
struc2vec	✗	✗[a]	✗[a]	✗	✗	✗	564 s
GraphWave	✗[c]	~	✓	✓	✗	✓	18 s
Role2Vec	~	✓[b]	✗	✗	~	✗	10 s
RESD	✗	✗	✗	✗	✗	✗	8 s
ReFeX	~	~	~	✓	~	✓	0.1 s
RWSE	✓	~	✓	✓	✓	✓	1.7 s

[a] struc2vec++ [5] extends it. [b] In the paper; the reference implementation is undirected. [c] Scale chosen from the spectrum and $1/n$ factor (Proposition 11). "No training": struc2vec and Role2Vec fit skip-gram models, RESD a variational autoencoder. [TODO: verify RESD, ReFeX, RWSE rows]

Table 2: Role classification accuracy. struc2vec is unseeded: mean ± std over 5 runs. The random-walk heat-kernel CF is RoleWalk's characteristic function on the rows of $\exp(s(P-I))$, $s \in \{1,2,3\}$, with the same frequencies: fixed scales and no $1/n$, but a random-walk rather than GraphWave's spectral heat kernel. Top-k + layers: the 20 largest probabilities at each step instead of the characteristic function, same layers, same dimension. Local statistics: see Table 8. `roles.csv`, `roles_gw.csv`, `roles_matched.csv`, `struc2vec_runs.csv`.

	Brazil	Europe	USA	Reality-call	Actor	Film*
RoleWalk	0.743	0.583	0.657	0.860	**0.487**	0.569
RoleWalk + curv.	0.741	0.596	**0.666**	0.860	0.485	0.581
struc2vec	0.705 ± .022	0.561 ± .020	0.596 ± .004	–	–	–
RESD	0.728	0.544	0.653	0.857	0.476	0.563
GraphWave	**0.769**	0.561	0.587	0.860	0.481	–
GraphWave, no $1/n$	**0.769**	0.560	0.587	0.860	0.481	–
Random-walk heat-kernel CF	0.750	**0.598**	0.633	0.859	0.484	0.549
Top-k + layers	0.672	0.590	0.625	0.858	0.482	0.564
Local statistics	0.716	0.584	0.616	**0.862**	0.486	0.517
ReFeX	0.691	0.576	0.622	0.851	0.478	0.579
Role2Vec	0.476	0.366	0.562	0.844	0.399	**0.743**
RWSE	0.711	0.572	0.517	0.855	0.363	0.450
Degree (baseline)	*0.756*	*0.554*	*0.567*	*0.858*	*0.476*	*0.478*

* A node-type task (Section 6.5). "–": too slow or out of memory. Paired tests over the 50 splits (Nadeau–Bengio corrected intervals, `paired_tests.csv`): on USA, RoleWalk + curv. beats every other method significantly except RESD; on Brazil, Europe, Reality-call and Actor no difference is significant either way (intervals of ±0.05 on Brazil and Europe, 131 and 399 nodes); on Film, Role2Vec wins. Appending the local statistics to RoleWalk gives 0.752 / 0.595 / 0.659 / 0.861 / 0.486 / 0.598.

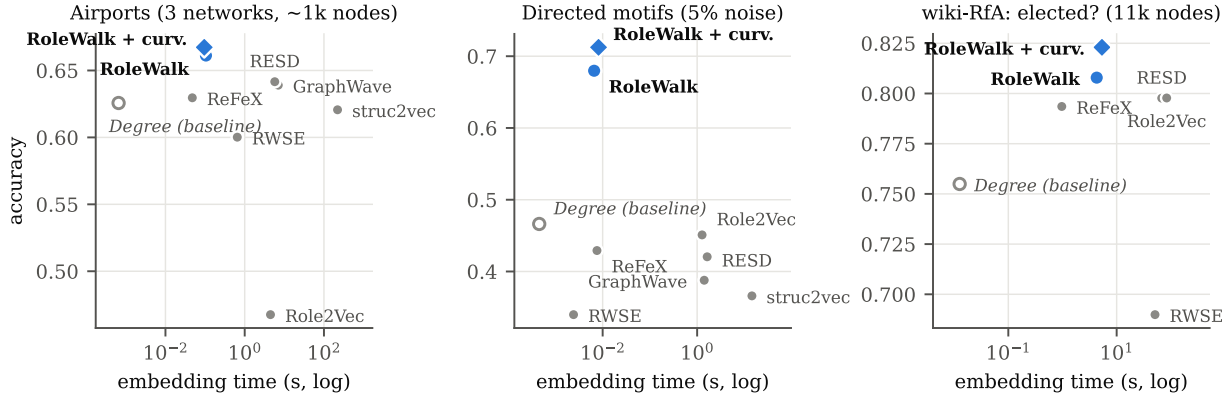


Figure 1: Accuracy against embedding time (one core, log scale), every method that runs on each benchmark. Airports: mean over Brazil, Europe and USA (struc2vec: mean of 5 runs). Directed motifs: 5% rewired edges, mean of 3 seeds. wiki-RfA: task A of Table 5. Hollow marker: degree, a trivial baseline rather than an embedding. RoleWalk is the only method in the top-left corner of every panel. `speed_figure.py`.

Table 3: Wikipedia talk networks (RONE): precision@10 for bots and admins, mean over the 6 networks, and over the 3 smallest (446–2,101 nodes), the only ones struc2vec runs on. `roles.csv`.

	Bots (6)	Admins (6)	Bots (3)	Admins (3)
RoleWalk	**0.129**	**0.324**	**0.179**	0.356
RoleWalk + curv.	0.121	0.322	0.162	0.356
struc2vec	–	–	0.146	**0.391**
GraphWave	0.115	0.298	0.163	0.341
RESD	0.114	0.254	0.158	0.334
ReFeX	0.101	0.299	0.161	0.375
Role2Vec	0.119	0.016	0.158	0.041
RWSE	0.118	0.297	0.178	0.309
Degree (baseline)	*0.075*	*0.296*	*0.121*	*0.341*

Mixed: RoleWalk leads on bots and on admins over all networks, struc2vec on admins where it runs. Networks are distributed undirected. [TODO: directed versions from SNAP]

Table 4: Directed roles on synthetic graphs (directed motifs hung from a directed cycle; several roles coincide once directions are dropped). 3 seeds; noise = share of edges rewired. Accuracy: logistic regression; ARI: k-means; mAP: retrieval. ReFeX, directed: in-, out- and mutual degrees and recursive features over out- and in-neighbours (450 one-hot features). `directed.csv`.

	Acc. 0%	Acc. 5%	Acc. 10%	ARI 5%	mAP 5%
RoleWalk + curv. + mixed	**1.000**	0.732	0.595	0.396	**0.629**
RoleWalk + mixed	**1.000**	0.728	0.590	0.373	0.615
RoleWalk + curv.	0.989	0.712	0.547	0.381	0.625
RoleWalk	0.908	0.680	0.510	0.366	0.574
RoleWalk, undirected	0.767	0.560	0.420	0.351	0.479
ReFeX, directed	**1.000**	**0.756**	**0.638**	0.355	0.607
RWSE, out + in	0.357	0.336	0.336	0.048	0.179
Top-k + layers	0.908	0.671	0.516	0.348	0.568
Local statistics (in / out)	0.863	0.675	0.584	**0.409**	0.607
Role2Vec	0.730	0.451	0.368	0.148	0.295
ReFeX	0.532	0.429	0.431	0.342	0.356
GraphWave	0.586	0.388	0.383	0.317	0.416
RESD	0.547	0.420	0.423	0.253	0.348
struc2vec	0.510	0.366	0.277	0.141	0.260
In/out degree (baseline)	*0.516*	*0.466*	*0.405*	*0.310*	*0.349*

Table 5: Real directed roles: Wikipedia requests for adminship [19] (11,377 users, 188k votes voter → candidate; signs dropped, repeated votes merged, unweighted). A: elected among the 3,494 candidates. B: voter / failed candidate / admin. `roles_rfa.csv`, `roles_rfa_new.csv`, `roles_rfa_mixed.csv`.

	A: elected	B: 3 classes	Time
RoleWalk + curv. + mixed	**0.827**	**0.944**	7 s
RoleWalk + mixed	0.814	0.940	7 s
RoleWalk + curv.	0.823	0.943	5 s
RoleWalk	0.808	0.938	4 s
RoleWalk, undirected	0.791	0.863	
RoleWalk + curv., undirected	0.814	0.900	
RESD	0.798	0.843	68 s
Role2Vec	0.798	0.857	82 s
ReFeX	0.794	0.847	1 s
ReFeX, directed	0.778	0.930	3 s
Top-k + layers	0.804	0.936	5 s
Local statistics (in / out)	0.814	0.941	3 s
RWSE (out-walk)	0.690	0.826	51 s
RWSE, out + in	0.710	0.836	84 s
In/out degree (baseline)	*0.755*	*0.922*	*0 s*

struc2vec and GraphWave do not run at this size.

Table 6: Expressiveness: pairs told apart (CSL: number of the 10 isomorphism classes told apart). Every set is 1-WL-indistinguishable. Two graphs are told apart when their multisets of node embeddings differ beyond float32 noise (sorted rows compared at a relative tolerance 10^{-4}). No false difference on random relabellings. `expressiveness.csv`.

	Atlas ≤ 7 (20)	Regular (3,181)	CSL (10 cl.)	Decalin / bicyc.	$2C_3$ / C_6	SRG pair*
RoleWalk (3 steps)	**20**	**3,181**	6	yes	yes	no
RoleWalk (5 steps)	**20**	**3,181**	**10**	yes	yes	no
RoleWalk, no layers	16	3,180	6	yes	yes	no
RWSE (16 steps)	**20**	3,180	**10**	yes	yes	no
Degree (baseline)	*0*	*0*	*1*	*no*	*no*	*no*

* Shrikhande / 4×4 rook: strongly regular with equal parameters, not separated by 2-FWL (classical 3-WL) nor by Proposition 7.

Table 7: BREC [14] (400 pairs).

	Basic (60)	Regular (100)	Extension (100)	CFI (100)	4-vertex (20)	Dist.-reg. (20)	Total
RoleWalk (3 steps)	60	50	97	0	0	0	207
RoleWalk (5 steps)	60	50	100	3	0	0	213
RoleWalk, no layers	58	48	97	0	0	0	203
RWSE (16 steps)	60	50	100	4	0	0	**214**

Distance-regular: 0 by Proposition 7. The 50 regular pairs missed are exactly the 50 strongly regular pairs, all with equal parameters (checked).

Table 8: Centrality transfer: a gradient-boosting regressor from features to centrality is fitted on one graph and evaluated (Spearman) on another graph of the same family and size (other seed), 6 families (Barabási-Albert, powerlaw-cluster, Watts-Strogatz, SBM, random geometric, motifs) × 2 seeds. Path-based centralities (top four rows) are where walks matter. ReFeX's feature selection, scaling and bins are fitted on the training graph and frozen. Ablation: RoleWalk with the local statistics appended. `centrality_transfer.csv`.

	RoleWalk	+ curv.	Top-k + layers	Local stats	ReFeX (frozen)	RW heat CF	RWSE	GraphWave, no $1/n$	*Degree*	*RW + local*
Betweenness	0.902	**0.905**	0.884	0.857	0.858	0.883	0.708	0.714	*0.634*	*0.910*
Closeness	**0.872**	0.868	0.817	0.752	0.766	0.809	0.732	0.598	*0.428*	*0.875*
Harmonic	**0.922**	0.921	0.879	0.820	0.832	0.874	0.784	0.695	*0.500*	*0.924*
PageRank	**0.990**	**0.990**	0.989	0.984	0.974	0.986	0.864	0.857	*0.927*	*0.991*
Clustering	0.874	0.871	0.870	**0.914**	0.902	0.862	0.803	0.400	*0.212*	*0.912*
k-core	0.746	0.749	**0.750**	0.748	0.727	0.742	0.660	0.707	*0.692*	*0.750*
Eigenvector	0.654	0.649	0.655	0.647	**0.693**	0.575	0.530	0.463	*0.428*	*0.655*
Mean	**0.851**	0.850	0.835	0.817	0.822	0.819	0.726	0.633	*0.546*	*0.859*

Local stats: degree, neighbour-degree moments, triangles, egonet boundary, clustering, lazy-walk returns at steps 2 and 3. With an unfrozen ReFeX (only where feature counts happen to match), k-core reached 0.935: an artefact of the subset. RW heat CF: random-walk heat-kernel characteristic function. GraphWave with its $1/n$ is within 0.006 of the no-$1/n$ column.

Table 9: Role transfer between airport networks: classifier fitted on one network, tested on another (6 ordered pairs, accuracy). Labels are activity quartiles, close to degree: this transfer is won by degree-level statistics, and the centrality transfer of Table 8 is the relevant test of what walks transfer. Removing GraphWave's $1/n$ lifts it from chance to 0.455 (Proposition 11); fixed scales and a heat kernel instead of walk powers give 0.425. `role_transfer.csv`, `role_transfer_gw.csv`.

RoleWalk	+ curv.	RoleWalk, no layers	Top-k + layers	Local statistics	ReFeX (frozen)
0.488	0.472	0.445	0.475	**0.565**	0.438
RWSE	GraphWave	GraphWave, no $1/n$	RW heat CF	*Degree (baseline)*	
0.375	0.249 (chance)	0.455	0.425	*0.542*	

Table 10: Structural encodings for GNNs on ZINC (test MAE at the best validation epoch). Top: GINE, 4 layers, ≈230k parameters, 300 epochs, 3 seeds. Bottom: GPS [9] protocol, 10 layers, ≈470k parameters, 500 epochs, 4 seeds. `pe_zinc_v13.csv`, `pe_zinc_gps*.csv`.

GINE	Test MAE	
none	0.202 ± 0.010	
RWSE	0.107 ± 0.004	
RoleWalk	0.108 ± 0.008	
RoleWalk + RWSE	0.104 ± 0.006	
RoleWalk + curv. + RWSE	**0.095 ± 0.002**	
GPS	Test MAE	Val. MAE
RWSE (published: 0.070 ± 0.004)	0.076 ± 0.004	0.099 ± 0.006
RoleWalk + curv.	0.075 ± 0.002	**0.092 ± 0.004**
RoleWalk + curv. + RWSE	**0.072 ± 0.002**	0.094 ± 0.006

RoleWalk + curv. + RWSE is below RWSE on each of the 4 seeds (paired difference 0.004 ± 0.004; 0.0002 on the closest seed). [TODO: 5th seed running; Peptides (LRGB)]

Table 11: Embedding time against size with the fused kernel, defaults (tol 10^{-4}, 3 steps, layers), including the conversion of the networkx graph. `scalability.csv`.

Nodes	10 k	30 k	100 k	300 k	1000 k
Barabási-Albert ($m = 3$), 1 core	0.47 s	1.84 s	8.94 s	43.05 s	192.93 s
+ curvature	0.56 s	1.94 s	11.22 s	52.36 s	246.86 s
8 threads	0.22 s	0.75 s	6.34 s	13.69 s	72.7 s
Watts-Strogatz ($k = 6$), 1 core	0.12 s	0.45 s	1.78 s	5.11 s	17.65 s
+ curvature	0.16 s	0.48 s	1.77 s	5.39 s	19.16 s
8 threads	0.06 s	0.31 s	0.81 s	2.44 s	9.78 s

Peak memory 2.5 GB at a million nodes, of which 1.6 GB hold the networkx graph. Baselines on the same Barabási-Albert graphs, one core (`scalability_baselines.csv`): ReFeX 0.1 s / 1.8 s / 31 s at 10k / 100k / 1M nodes, faster than RoleWalk; 16-step RWSE 68 s at 10k and over 16 GB at 100k; the heat-kernel CF 196 s at 10k. At 1,190 nodes (USA airports): struc2vec 564 s, GraphWave 18 s, Role2Vec 10 s, RESD 8 s; struc2vec and GraphWave do not run on wiki-RfA (11k nodes).

6 Analysis

6.1 Summaries of the walk distribution

Table 12: Same budget of features, replacing the characteristic function. Airports: mean accuracy; Transfer: airport transfer; Motifs: synthetic accuracy.

	Airports	Transfer	Motifs (varied / houses)	Time 20k
Characteristic function	0.656	0.445	**0.744** / 0.733	6.5 s
Laplace transform	**0.657**	0.345	0.614 / 0.717	18 s
Dyadic histogram (Haar)	0.631	0.353	0.614 / 0.725	15 s
Power sums (Rényi)	0.653	0.420	0.647 / 0.692	9.7 s
Top-10 probabilities	0.652	**0.512**	0.679 / **0.758**	3.5 s

6.2 Geometry: moments rather than integrated characteristic functions

Table 13: Ways to add local geometry (ablation runs: may differ from the main tables in the third decimal). Here, degree marks and counting bounds hurt transfer.

	Airports	Transfer	Motifs (varied / houses)	Directed acc.
No geometry	0.656	0.445	0.744 / 0.733	0.678
+ onion layers (moments)	0.661	0.488	0.744 / 0.758	0.682
+ layers, one CF per layer	0.654	0.424	0.751 / 0.742	0.695
+ degree marks (joint CF)	0.65–0.67	0.34–0.36	–	–
+ curvature, counting bound	0.667	0.388	**0.754** / **0.767**	0.721
+ curvature, probabilistic (moments)	**0.668**	0.470	0.725 / 0.742	0.712
+ curvature, exact Ollivier-Ricci	0.662	**0.494**	0.746 / 0.750	**0.743**
+ curvature, integrated CF	0.663	0.400	0.751 / 0.750	0.697

The probabilistic bound is kept over exact Ollivier-Ricci for cost (no optimal transport) at a small loss.

Marks on the steps of a path rather than on its end were also tried: the parity of steps inside a layer (closing odd cycles), obtained from one extra signed walk $\tilde{D}^{-1}(A - I)$ and the layer of arrival, since on undirected graphs $\#\text{intra} \equiv \#\text{moves} - d(u, j) \pmod 2$; its exact count; and the order of moves out and in, as products in the quaternion group Q_8. On the 6,781 nodes of connected graphs with at most 7 nodes, none separates a pair of nodes that X does not, and none improves airport accuracy (`intra_layer_study.py`, `quaternion_study.py`). With labels g (out) and g^{-1} (in) in an abelian group and trivial labels otherwise, the product along a path only depends on its end layer, which gives back the layers; short cycles already change the multiset of probabilities.

A purer object was also tried: one characteristic function of each node's whole trajectory $\left(1[j = u], p^1_{uj}, p^2_{uj}, p^3_{uj}\right)$, sampled along marginal and mixed-time projections (walks stopped at a random time), in place of per-step functions and layers. It sees whether large probabilities at different steps sit on the

same node, which per-step multisets forget, and makes every coordinate Lipschitz. At 48 to 160 features it separates the same small-graph nodes as RoleWalk and matches it on airports (0.664–0.666 against 0.662), but transfers worse (0.454–0.471 against 0.486) and loses on directed motifs (ARI 0.295–0.314 against 0.366) (`trajectory.csv`): hard layers are a better use of a few dimensions.

6.3 Pruning

On the six real networks, pruning at the default tol 10^{-4} changes accuracy by at most 0.005 (0.002 with a budget), and by at most 0.012 at tol 10^{-3} (`pruning.csv`). Without a budget, the characteristic-function block keeps a median relative error below 10^{-4} at tol 10^{-4}, but layers move by up to 16% (73% at tol 10^{-3}). The error concentrates on hubs: for the centre of a star with 20,000 leaves and the largest hub of a 100k-node Barabási-Albert graph, the embedding is off by 49% and 27%. A budget δ caps the worst layer error at δ (0.009, 0.047, 0.096 for $\delta = 0.01, 0.05, 0.10$) and makes both hubs exact; the observed error is at most 0.51 of the certificate of Proposition 13 everywhere. On the real networks the budget costs nothing measurable (3.3 s against 3.1 s in total); on the hub-heavy Barabási-Albert graph it doubles the time, as hub rows are kept whole.

6.4 Kernel scaling

In raw embeddings $\theta = 100$ carries 53% of the variance; unscaled layers carry 0.02% and leave k-means roles unchanged. A grid of 48 configurations (frequency weighting, range, steps, layers), scored on synthetic and real networks with equal weight (`kernel_grid.csv`): scaling the layers by $\|\Theta\|$ is what matters (score 0.274 $\rightarrow$ 0.323), the frequency weighting barely does, and the defaults win whichever dataset is left out. Spacing matters less than range (`theta_scheme.csv`): on $[1, 100]$, log-spaced frequencies score 0.323 against 0.316 for uniform ones, with the same airport accuracy (0.661 / 0.662) and transfer (0.488 / 0.485). A narrower range helps real networks and transfer but loses on synthetic motifs, whatever the spacing: $[1, 30]$ gives transfer 0.515 (log or uniform) and score 0.312–0.315; log $[1, 50]$ gives 0.506 and 0.326, within noise of the default. Log spacing is kept as a heuristic: probabilities spread over several decades, which a log grid samples evenly, while a uniform grid puts 9 of its 10 frequencies above 10. Feature-wise standardisation, as in our classifiers, cancels any constant factor on a block: the scaling of the layers matters for raw distances (k-means, Ward, nearest neighbours), not for the classification tables.

6.5 Film is a node-type task

Table 14: Film (RONE). Class 3 is an independent set in a separate block of node ids.

	Class 3 vs rest	Within classes 0–2
Role2Vec	**0.946**	0.637
RoleWalk + curv.	0.740	<u>0.634</u>
ReFeX	<u>0.745</u>	0.627
Degree (baseline)	*0.536*	*0.623*
Majority class	0.593	0.623

No method predicts classes 0–2 clearly above the majority class; edge homophily is 0.28, below the 0.32 expected from random labels with the same (degree-weighted) class proportions. Role2Vec's lead on class 3 is consistent with walk co-occurrence, i.e. position, a hypothesis we have not tested (a node-ID permutation check would); we read Film as a node-type task rather than a role benchmark.

7 Limitations

- Walk statistics cannot separate distance-regular graphs with equal arrays (Proposition 7), most CFI pairs (3 of 100 with 5 steps), or graphs whose balls coincide (locality); without curvature, some nodes separated by 1-WL are confused (Proposition 9).
- Without a mass budget, the pruning certificate is only a posteriori and large on hubs (Section 6.3); the budget costs time on hub-heavy graphs, and has no support cap.
- ReFeX is faster (5× at a million nodes) and wins on noisy synthetic directed motifs in supervised accuracy and on k-core transfer; a fixed-scale heat-kernel CF matches RoleWalk on two of three airport networks.

- Middle-tier roles on real networks often form a continuum; per-node stability should be reported next to roles.
- Degree-like labels (airports) favour trivial baselines.
- One real labelled directed benchmark.
- GPS on ZINC: 4 seeds; our RWSE reproduction (0.076) is weaker than the published 0.070; RoleWalk + curvature + RWSE (0.072) is better on every seed, by a margin of the size of the seed noise.

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